Docking and scoring notebook

Protocol for docking ligands into a target protein and rescoring the poses:

1. **RDKit** generates a diverse, low-energy set of conformations for each ligand.
2. The binding site is defined either from a **protein–ligand complex PDB** (the ligand is extracted for you) or from **prepared files** (protein without ligand + extracted ligand).
3. **smina** (https://sourceforge.net/projects/smina/) docks the conformers into the site (rigid or flexible side chains).
4. **AEV-PLIG** (Valsson *et al.*, *Commun. Chem.* 2025) predicts the binding affinity (pK) of every pose.

The target is the **Zika virus NS2B-NS3 protease**, docking the nine-ligand `Trial10.sdf` trial set.

Kernel: `docking_scoring` conda env. One-off setup, in a terminal in the project folder (see README.md for details and for installing smina):

```
conda env create -f environment.yml
conda run -n docking_scoring python -m ipykernel install --user --name docking_scoring --display-name
"Python (docking_scoring)"
conda run -n docking_scoring python aev_plig/download_models.py
```

Output files are written next to the input files they come from; all prediction results go to a `predictions` folder next to the docked SDF.

Widgets: each step has a panel with a **Run** button. Set `AUTO_RUN = True` in the next cell to run every step with the default settings when executing all cells.

Docking_scoring — dock a ligand series and rescore the poses

RDKit conformers → smina docking → AEV-PLIG affinity prediction, with the crystal ligand as the benchmark

INPUTS

Ligands to dock
SDF of 3D structures; different molecules must not share a name (checked)
RDKit

Protein–ligand complex
PDB or mmCIF. All ligands are listed, largest preselected; the chosen one defines the site
gemmi · RDKit · RCSB CCD

or

…or prepared files
a protein with no ligand plus the ligand extracted from the site, as in the old notebook
either route works

NOTEBOOK STEPS

the series you want to dock — *either route — step 3 produces the receptor and the box*

1b Ionise
predict pKa and set the state at pH 7.4 (optional)
QupKake · GFN2-xTB

2 Conformers
srETKDGv3, MMFF94, RMSD-pruned, energy window
RDKit

3 Binding site
split receptor, extract ligand, size the box, ray-traced images
docking_tools · PyMOL

4 Dock
rigid, flexible side chains, or minimise in place
smina

5 Browse poses
step through with a slider — 2D structure beside the 3D view
py3Dmol · RDKit

6 Score
binding affinity as pK, plus ligand efficiency and the PDB ligand
AEV-PLIG ×10

7 Export
best pose per ligand and a session to open in PyMOL
PDB · SDF · .pse

checked as it runs: every ligand fits the search box · no ligand silently dropped by smina · each pose paired with its own receptor (flexible runs)

OUTPUTS — written beside the inputs they came from

Docked poses
<input>_docked.sdf, the smina log and a run record (.run.json)
re-runnable

predictions/
per-pose summary and the best-pose table: pK, ligand efficiency, box margin
CSV

Best poses
ligand-only and protein–ligand complex PDB, plus SDF
one per ligand

Pictures
PyMOL session, binding-site images, smina-vs-pK plot
.pse · .png

Reproducibility:

each run records the smina version, the full command, the seed, every setting, the platform and SHA-256 of its inputs, so a result can be traced back to exactly how it was produced.

Metal sites: AEV-PLIG reads only the standard amino acids, so a metal coordinating the ligand is invisible to it and smina has no metal term — treat scores for metalloenzymes with caution.

In [1]:
```
import os
from pathlib import Path

import pandas as pd
from IPython.display import HTML, display
from rdkit import Chem
from rdkit.Chem import Draw

import docking_tools as dt

%matplotlib inline

AUTO_RUN = os.environ.get("DOCKING_AUTORUN") == "1"   # set True to run all steps with defaults
dt.PROTEIN_COLOURING = "rasmol"   # protein cartoons: "rasmol" (secondary structure), "rainbow" or "grey"

print(dt.cpu_note())
dt.library_versions()
```

```
24 cores available (16 performance + 8 efficiency)
```

Out[1]:

	old env (docking)	this env
python	3.8.19	3.12.14
rdkit	2024.3.5	2026.09.1
torch	2.4.1	2.10.0
torch_geometric	2.6.1	2.8.0.post1
torchani	2.2.4	2.9.0
py3Dmol	2.5.4	2.5.4
pandas	2.0.3	3.0.6
numpy	1.24.4	2.5.3
scipy	1.10.1	1.18.1
sklearn	1.3.2	1.9.1
biopandas	0.5.1	0.5.2
qcelemental	0.30.1	0.51.2
ipywidgets	8.1.5	8.1.9
matplotlib	3.7.3	3.11.2
pymol	-	3.1.0
openbabel	-	3.2.1

1. Ligands to dock

Set the path of the SDF file of structures you want to dock. The name must be on the first line of each record, e.g.

```
AEM 10028511
MOE2019           2D
```

Conformers, docked poses and predictions will be written to the **same folder** as this file.

> The nine molecules in `Trial10.sdf` are titled with their **SMILES strings** rather than names, so the check below offers the SDF's `Molecule_Name` field (`blind_challenge_001` ...) as the ligand ID - take it, because the name ends up in pose file names, the results tables and the plot labels.
>
> `openbind-blind-challenge-1-TEST-BLINDED.sdf` in this folder is the full **400-molecule** set, in the same format; point `LIGAND_SDF` at it to run the whole challenge. Expect a long job - at exhaustiveness 20 the docking alone scales to many hours, so run it from a terminal rather than interactively.

In [2]:
```
# DOCKING_LIGANDS overrides the path, so the same notebook can run another set
LIGAND_SDF = Path(os.environ.get("DOCKING_LIGANDS", "Trial10.sdf")).resolve()

input_mols = dt.load_sdf(LIGAND_SDF)   # warns about unreadable or unnamed records
print(f"{len(input_mols)} molecules read from {LIGAND_SDF}")
```

```
9 molecules read from Trial10.sdf
```

Check ligand names

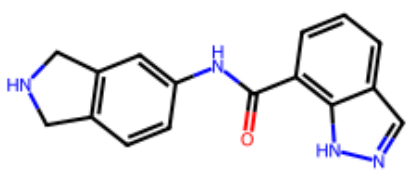
smina keeps only the molecule name, and the results (best pose per ligand, PDB export, plots) are grouped by name, so **different structures must not share a name**. If duplicates are found you can either use an **identifier field** from the SDF (e.g. `ID` , `IDNUMBER` , `Record_ID` – offered only if every molecule has a unique value) as the ligand ID, or have **unique names assigned automatically** (`name` , `name_2` , `name_3` ...). The original name is kept in the `original_name` property. The input SDF itself is not changed. Conformer generation is blocked until duplicates are resolved.

In [3]:
```
name_check = dt.LigandNamePanel(input_mols)
display(name_check.widget())
if AUTO_RUN and not name_check.resolved:
    name_check.apply()   # default: first unique identifier field, otherwise automatic names
```

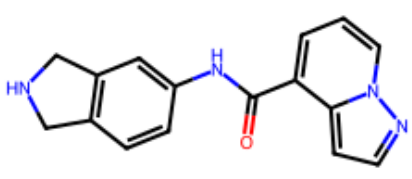
```
Ligand names now from SD field 'Molecule_Name' (original names kept in 'original_name'):
  blind_challenge_001, blind_challenge_002, blind_challenge_003, blind_challenge_004, blind_challenge_005,
blind_challenge_006, blind_challenge_007, blind_challenge_008, blind_challenge_009
```

In [4]:
```
# View as grid (re-run after applying a naming choice above)
Draw.MolsToGridImage([dt.to_2d(m) for m in input_mols], molsPerRow=4, subImgSize=(250, 200),
                     legends=[m.GetProp("_Name") for m in input_mols])
```

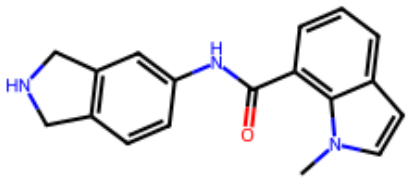
Out[4]:



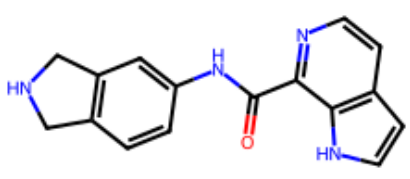
blind_challenge_001



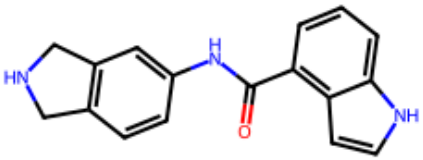
blind_challenge_002



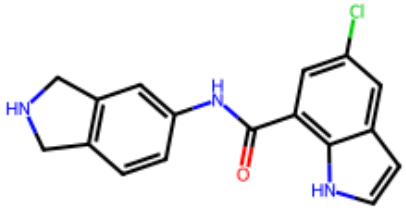
blind_challenge_003



blind_challenge_004



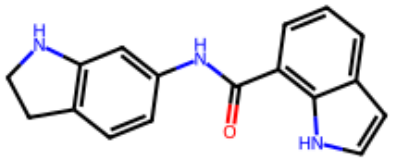
blind_challenge_005



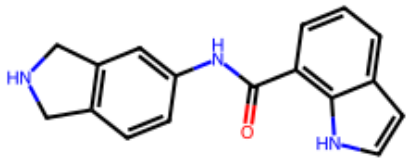
blind_challenge_006



blind_challenge_007



blind_challenge_008



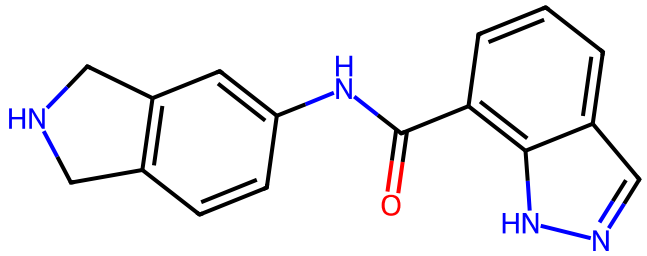
blind_challenge_009

Browse individual input structures

Use the slider (or Prev / Next) to step through the molecules.

In [5]:
```
dt.molecule_browser(input_mols)
```

```
1/9  blind_challenge_001   heavy atoms: 21   SMILES: O=C(Nc1ccc2c(c1)CNC2)c1cccc2cn[nH]c12
```



1b. Ionisation states with QupKake

QupKake (Abarbanel & Hutchison, *J. Chem. Theory Comput.* 2024) predicts **micro-pKa** values by combining GFN2-xTB calculations with graph neural networks. Each ligand is neutralised, QupKake predicts the pKa of every acidic and basic site, and the ionisation state at the chosen pH is set:

- the **most basic site** is protonated if its pKa is above the pH, and the **most acidic site** is deprotonated if its pKa is below it (at most one of each; other sites that also cross the pH are listed in *also_crossing_pH*);
- **implausible basic sites are ignored** (on by default): QupKake sometimes reports sulphonamide N, amide N or aryl-amine N (e.g. the N-aryl nitrogen of an aryl-piperazine) as the most basic site, which would put the charge in the wrong place – these are listed in *ignored*.

The ligands in their predicted states are written to `<input>_pH7.4.sdf` next to the input, and – with *Dock the ligands in their predicted ionisation states* ticked – that file is used for conformer generation and docking, so all later outputs are named `<input>_pH7.4_…` and never overwrite the results for the input states. The same prediction is made for the ligand from the PDB (section 3), which is used as the reference in the final table.

QupKake runs in its own conda environment (`qupkake` , with xtb 6.4.1); if it is not installed the input states are used. **Re-run the conformer cell after running QupKake.**

In [6]:
```
prot_panel = dt.ProtonationPanel(input_mols, LIGAND_SDF)
display(prot_panel.widget())
if AUTO_RUN and dt.qupkake_available():
    prot_panel.run()
```

```
Running QupKake (GFN2-xTB + GNN); about 30 s for 10 molecules ...
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
[20:18:55] Running Uncharger
Ionisation states written to Trial10_pH7.4.sdf (table: Trial10_pH7.4.csv; QupKake files in qupkake/Trial10)
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
[20:19:17] Running Uncharger
```

	name	changed	input_charge	predicted_charge	protonated	deprotonated	ignored	also_crossing_pH	input_smiles	smiles
0	blind_challenge_001	True	0	1	b:N11=7.44				O=C(Nc1ccc2c(c1)CNC2)c1cccc2cn[nH]c12	O=C(Nc1ccc2c(c1)C[NH2+]C2)c1cccc2cn[nH]c12
1	blind_challenge_002	True	0	1	b:N11=7.63				O=C(Nc1ccc2c(c1)CNC2)c1cccn2nccc12	O=C(Nc1ccc2c(c1)C[NH2+]C2)c1cccn2nccc12
2	blind_challenge_003	True	0	1	b:N20=7.71				Cn1ccc2cccc(C(=O)Nc3ccc4c(c3)CNC4)c21	Cn1ccc2cccc(C(=O)Nc3ccc4c(c3)C[NH2+]C4)c21
3	blind_challenge_004	True	0	1	b:N11=7.65				O=C(Nc1ccc2c(c1)CNC2)c1nccc2cc[nH]c12	O=C(Nc1ccc2c(c1)C[NH2+]C2)c1nccc2cc[nH]c12
4	blind_challenge_005	True	0	1	b:N11=7.62				O=C(Nc1ccc2c(c1)CNC2)c1cccc2[nH]ccc12	O=C(Nc1ccc2c(c1)C[NH2+]C2)c1cccc2[nH]ccc12
5	blind_challenge_006	True	0	1	b:N11=7.55				O=C(Nc1ccc2c(c1)CNC2)c1cc(Cl)cc2cc[nH]c12	O=C(Nc1ccc2c(c1)C[NH2+]C2)c1cc(Cl)cc2cc[nH]c12
6	blind_challenge_007	True	0	1	b:N16=7.63				Cc1ccc(C(=O)Nc2ccc3c(c2)CNC3)c2[nH]ccc12	Cc1ccc(C(=O)Nc2ccc3c(c2)C[NH2+]C3)c2[nH]ccc12
7	blind_challenge_008	False	0	0			b:N10=4.77 (aryl-amine N)		O=C(Nc1ccc2c(c1)NCC2)c1cccc2cc[nH]c12	O=C(Nc1ccc2c(c1)NCC2)c1cccc2cc[nH]c12
8	blind_challenge_009	True	0	1	b:N11=7.60				O=C(Nc1ccc2c(c1)CNC2)c1cccc2cc[nH]c12	O=C(Nc1ccc2c(c1)C[NH2+]C2)c1cccc2cc[nH]c12

2. Conformation generation

From the RDKit documentation: conformer generation is a difficult and subtle task. smina rotates around torsions during docking, but it does not flip rings or change amide geometry, so supplying several ring/amide conformers helps.

Are there better alternatives within RDKit than the original cell? Yes. The original notebook called `EmbedMultipleConfs(m, numConfs=3)` with no pruning or minimisation, so it could return near-duplicate, unrelaxed geometries. This notebook uses:

Step	What it does	Why
srETKDGv3 (default) / ETKDGv3 / ETKDG / ETDG	distance geometry + experimental torsion preferences	v3 has improved torsion libraries and macrocycle handling; the *sr* variant adds small-ring torsion preferences (better 3–7 membered ring puckers)
`pruneRmsThresh`	discards embeddings closer than the RMS threshold	diversity rather than duplicates
MMFF94 optimisation (UFF fallback)	relaxes bond lengths/angles and intramolecular clashes	cleaner starting geometry for docking
Energy window	drops conformers more than *n* kcal/mol above the minimum	removes strained geometries
Post-optimisation RMS de-duplication (symmetry-aware heavy-atom RMSD)	minimisation often collapses conformers onto each other	keeps only distinct ones
Optional stereoisomer enumeration	expands unassigned stereocentres	avoids docking an arbitrary isomer

Other RDKit options not used here: Torsion-Fingerprint/Butina clustering for very large ensembles, and `rdDistGeom.EmbedMultipleConfs` with `useRandomCoords=True` (used automatically as a fallback when embedding fails). For a single rigid molecule set *Conformers to embed* = 1.

The output file is `<input>_confs.sdf` in the input folder.

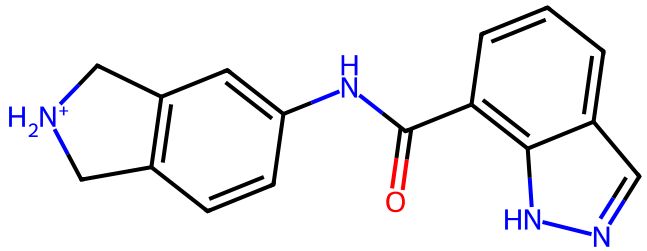
In [7]:
```
DOCK_MOLS, DOCK_SDF = prot_panel.docking_input()   # predicted ionisation states if QupKake was run
print(f"Docking {len(DOCK_MOLS)} ligands from {DOCK_SDF.name}")
conf_panel = dt.ConformerPanel(DOCK_MOLS, DOCK_SDF)
display(conf_panel.widget())
if AUTO_RUN:
    conf_panel.run()
```

```
Docking 9 ligands from Trial10_pH7.4.sdf
Wrote 60 conformers to Trial10_pH7.4_confs.sdf
```

	name	embedded	kept	force field
0	blind_challenge_001	17	8	MMFF94
1	blind_challenge_002	18	8	MMFF94
2	blind_challenge_003	15	5	MMFF94
3	blind_challenge_004	16	2	MMFF94
4	blind_challenge_005	15	8	MMFF94
5	blind_challenge_006	18	7	MMFF94
6	blind_challenge_007	18	7	MMFF94
7	blind_challenge_008	18	8	MMFF94
8	blind_challenge_009	18	7	MMFF94

In [8]:
```
# Check the conformer file (re-run this cell after generating)
CONF_SDF = conf_panel.out_path
if CONF_SDF.exists():
    confs = dt.load_sdf(CONF_SDF)
    print(f"{len(confs)} conformers in {CONF_SDF}")
    dt.molecule_browser(confs)
```

```
60 conformers in Trial10_pH7.4_confs.sdf
1/60  blind_challenge_001   heavy atoms: 21   SMILES: O=C(Nc1ccc2c(c1)C[NH2+]C2)c1cccc2cn[nH]c12
```



3. Protein and binding site

Two options:

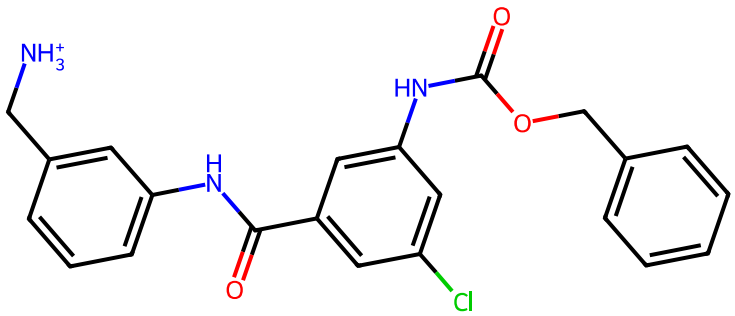
- **Complex structure** – a protein–ligand complex in **PDB or mmCIF** format (`.pdb` , `.cif` , `.mmcif` , optionally `.gz` ; mmCIF is now the PDB's standard format, e.g. a `.cif` downloaded from RCSB). All **ligands** are listed with their chemical names and the **largest is selected automatically** – choose another from the list if the file contains several ligands. Waters, solvents and crystallisation additives (e.g. DMSO, glycerol, sulfate), ions and terminal caps are hidden; tick *Also list solvents, ions, additives and caps* to show them. Only the first alternate location (A) is used. The ligand's bond orders come from (in order): an optional **SMILES template** (its charges are kept, so it can set the protonation state); the file's **hydrogens** and formal charges; the ligand's **PDB Chemical Component Dictionary** entry from RCSB (for files without hydrogens such as RCSB X-ray structures – neutral form); or **Open Babel** perception. Two files are written next to the input: `<stem>_protein.pdb` (protein only, no ligand/other HETs) and `<stem>_ligand_<RES><chain><num>.pdb/.sdf` . For mmCIF input `<stem>` ends in `_cif` (e.g. `ZIKV_NS2B-NS3_protease_cif_protein.pdb`) so it never overwrites files made from a PDB of the same entry, which may use a different coordinate frame.
- **Prepared files** – as in the original notebook: a protein PDB with no ligand plus the ligand extracted from the binding site (PDB).

> This structure is MOE-prepared, with hydrogens on both the protein and the ligand, so the ligand's bond orders and its **+1 charge** come straight from the file. Its residue code `01` is a local code rather than a PDB Chemical Component Dictionary one (the MOE header maps it to `A1B89`), which does not matter here because the hydrogens are present - no CCD lookup is needed. The bound ligand is the cationic benzamide `[NH3+]Cc1cccc(NC(=O)c2cc(Cl)cc(NC(=O)OCc3ccccc3)c2)c1` .
>
> **Keep both chains.** The receptor written by this step contains chain **A** (the NS2B cofactor) and chain **B** (the NS3 protease domain), and both are needed: of the 17 residues within 5 A of the ligand, 3 are from NS2B (Ser81, Gly82, Asp83) and 14 from NS3. The NS2B cofactor peptide completes the substrate groove, so docking into NS3 alone would be wrong.

Press **Use this binding site** after changing anything.

In [9]:
```
site = dt.BindingSiteSetup(complex_pdb=Path("ZIKV_NS2B-NS3_protease.pdb").resolve(),
                           protein_pdb=Path("ZIKV_NS2B-NS3_protease_protein.pdb").resolve(),
                           ligand_pdb=Path("ZIKV_NS2B-NS3_protease_ligand_01B203.pdb").resolve())
display(site.widget())
site.apply();   # applies the defaults; press the button to re-apply after changes
```

```
Protein for docking : ZIKV_NS2B-NS3_protease_protein.pdb  (2919 ATOM records)
Ligand (PDB)        : ZIKV_NS2B-NS3_protease_ligand_01B203.pdb
Ligand (SDF)        : ZIKV_NS2B-NS3_protease_ligand_01B203.sdf
Bond orders: RDKit DetermineBondOrders (net charge +1)
SMILES: [NH3+]Cc1cccc(NC(=O)c2cc(Cl)cc(NC(=O)OCc3ccccc3)c2)c1
```



01_B203

In [10]:
```python
# Ionisation state of the ligand from the PDB (crystal coordinates kept), with the same QupKake settings
REF_MOL = site.ligand_mol
if site.ligand_mol is not None and prot_panel.use.value and prot_panel.ionised is not None:
    REF_MOL, ref_table = dt.ionise_reference_ligand(site.ligand_mol, LIGAND_SDF.parent / "qupkake" / "pdb_ligand",
                                                    prot_panel.ph.value, prot_panel.filter.value)
    display(ref_table[["name", "changed", "input_charge", "predicted_charge", "protonated", "ignored", "smiles"]])
print("Reference ligand:", Chem.MolToSmiles(Chem.RemoveHs(REF_MOL)) if REF_MOL is not None else None)
```

```
[20:19:17] Running Uncharger
[20:19:17] Removed positive charge.
[20:19:46] Running Uncharger
[20:19:46] Removed positive charge.
```

	name	changed	input_charge	predicted_charge	protonated	ignored	smiles
0	01_B203	False	1	1	b:N29=8.37		[NH3+]Cc1cccc(NC(=O)c2cc(Cl)cc(NC(=O)OCc3ccccc...

```
Reference ligand: [NH3+]Cc1cccc(NC(=O)c2cc(Cl)cc(NC(=O)OCc3ccccc3)c2)c1
```

Images of the complex and the binding site

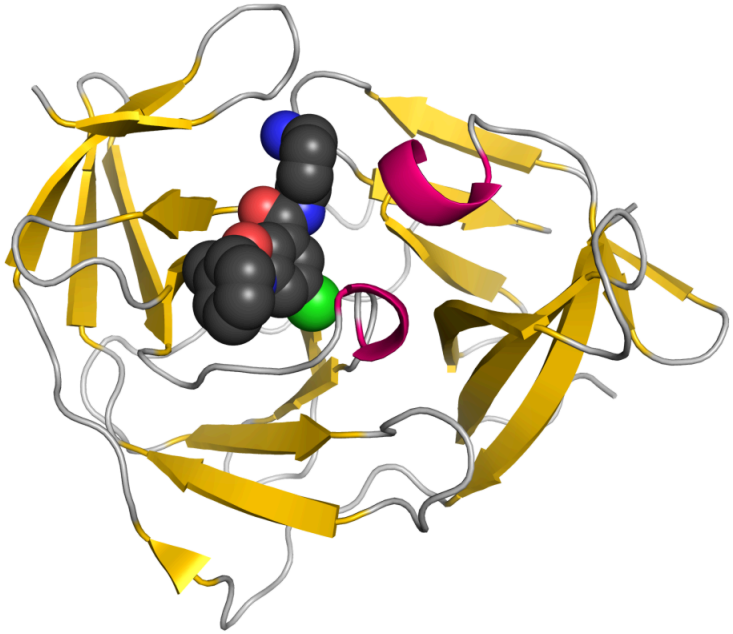
The protein is coloured by secondary structure with the **RasMol** scheme (helix magenta-red, sheet yellow, coil/turn white – drawn light grey so it shows on a white background); change `dt.PROTEIN_COLOURING` in the first code cell for rainbow or grey. Ray-traced with PyMOL and saved as `<stem>_complex.png` and `<stem>_site.png` next to the input structure. In the binding-site image residues within 5 Å of the ligand are shown as sticks, and polar contacts as orange dashes. Untick **Show binding-site side chains** to redraw the binding-site image without the side chains (and their polar contacts); the residues stay labelled.

In [11]:

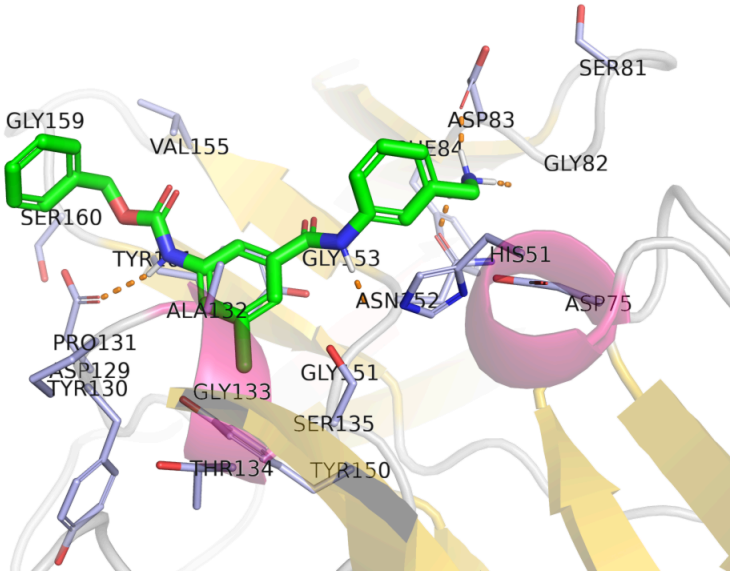
```
show_side_chains = dt.widgets.Checkbox(value=True, description="Show binding-site side chains",
                                       layout=dt.widgets.Layout(width="400px"))
image_out = dt.widgets.Output()

def _draw_images(_=None):
    global complex_png, site_png
    image_out.clear_output()
    with image_out:
        complex_png, site_png = dt.make_images(site.protein, site.ligand_ref, site.image_prefix,
                                               site_side_chains=show_side_chains.value)
        dt.show_images_side_by_side([complex_png, site_png],
                                    ["Protein with ligand bound", "Ligand in the binding site"])
        display(HTML(dt.colour_legend_html()))

show_side_chains.observe(_draw_images, names="value")
display(dt.widgets.VBox([show_side_chains, image_out]))
_draw_images()
```



Protein with ligand bound



Ligand in the binding site

Protein (RasMol secondary structure): helix sheet coil/turn

4. Docking with smina

The Zika virus NS2B-NS3 protease cleaves the viral polyprotein and is required for replication, which makes it a long-standing antiviral target (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6163217/). It is a two-component enzyme: the NS3 protease domain (chain B here) carries the catalytic triad **His51-Asp75-Ser135**, and a ~40-residue cofactor peptide from NS2B (chain A) wraps around it and completes the substrate-binding groove. The site is shallow, polar and strongly prefers basic P1/P2 groups, so most known inhibitors - including the ligand bound here and every ligand in the trial set - are **cationic**, which is why the pH 7.4 ionisation step in section 1b matters for this target.

The panel is pre-filled with the settings from the original notebook (`--exhaustiveness 20 --seed 0 --autobox_ligand`) and `--cpu` set to the number of cores detected. Notes:

- `--minimize` performs only a local energy minimisation of the supplied coordinates (no search). Use it on poses that are already in the site, e.g. the crystal ligand SDF or a docked SDF — not on freshly generated conformers. The output defaults to `<input>_minimized.sdf`.
- **Docking mode** – *Rigid receptor* (default) or *Flexible side chains*. ⚠ **Flexible docking is much slower** (typically 10–100× longer, rising steeply with the number of flexible residues). Residues can be chosen
  - **automatically** – every residue with an atom within a user-defined distance of the site-defining ligand (default 5 Å), or
  - **manually** – pick residues from the list (nearest first) and/or type extra ones (`A:41,A:145`).

  Gly, Ala and Pro are excluded (no side-chain torsions). smina then also writes `<output>_flexres.pdb` holding the flexed side chains for **every pose, in the same order as the docked SDF** – the viewer and AEV-PLIG use it to pair each ligand pose with its own receptor conformation.
- **Search-box check.** The box is the autobox ligand's extent (all atoms, hydrogens included) plus *autobox_add* on every side; smina only searches inside it and gives no warning if a ligand is too big. The panel compares each ligand's longest span (from the conformer file) with the box and warns when a ligand is *tight* (less than 2 Å to spare along the smallest box side) or *longer than the smallest box side*, suggesting an *autobox_add*. After docking, poses with a heavy atom within 1 Å of a box face are reported, and the results tables get `box_margin` (Å; negative = outside) and `at_box_edge` – a best pose at the edge means the box may be limiting that ligand.
- The docking log is written next to the output as `<output>.log`, together with a run record `<output>.run.json` (smina version, full command, seed, `--cpu` and all settings, platform, SHA-256 of the inputs) so every run is self-describing. With this smina build

the same seed gives the same poses for any `--cpu` (see README, Reproducibility).

In [12]:
```
panel = dt.SminaPanel(receptor=site.protein, autobox_ligand=site.ligand_pdb, ligands=CONF_SDF)
display(panel.widget())
if AUTO_RUN:
    # DOCKING_FLEXIBLE=<distance in A> runs flexible side chains instead of a rigid receptor,
    # choosing every flexible-capable residue within that distance of the site-defining ligand.
    if os.environ.get("DOCKING_FLEXIBLE"):
        panel.mode.value = "Flexible side chains (slower)"
        panel.flex_dist.value = float(os.environ["DOCKING_FLEXIBLE"])
        panel.find_flex_residues()
    panel.run()
```

```
$ /usr/local/smina.osx.12 -r ZIKV_NS2B-NS3_protease_protein.pdb -l Trial10_pH7.4_confs.sdf -o Trial10_pH7.4_docked.sdf
--seed 0 --cpu 24 --autobox_ligand ZIKV_NS2B-NS3_protease_ligand_01B203.pdb --autobox_add 4.0 --exhaustiveness 20 --
num_modes 9 --energy_range 3.0
# smina  master:dc3dfab+   Built Dec  1 2021.  Based on AutoDock Vina 1.1.2.
# mode rigid  seed 0  cpu 24  exhaustiveness 20  scoring vina
# platform Darwin arm64  started 2026-10-21T20:19:53+01:00
# run record: Trial10_pH7.4_docked.run.json
✔ Box check (before docking): box 17.7 x 17.7 x 23.1 A (7,230 A^3); longest ligand 12.6 A, smallest box side 17.7 A.
All ligands fit with >= 2 A to spare.

   _______  _______ _________ _        _______
  (  ____ \(       )\__   __/( (    /|(  ___  )
  | (    \/| () () |   ) (   |  \  ( || (   ) |
  | (_____ | || || |   | |   |   \ | || (___) |
  (_____  )| |(_)| |   | |   | (\ \) ||  ___  |
        ) || |   | |   | |   | | \   || (   ) |
  /\____) || )   ( |___) (___| )  \  || )   ( |
  \_______)|/     \|\_______/|/    )_)|/     \|
smina is based off AutoDock Vina. Please cite appropriately.
Weights      Terms
-0.035579    gauss(o=0,_w=0.5,_c=8)
-0.005156    gauss(o=3,_w=2,_c=8)
0.840245     repulsion(o=0,_c=8)
-0.035069    hydrophobic(g=0.5,_b=1.5,_c=8)
-0.587439    non_dir_h_bond(g=-0.7,_b=0,_c=8)
1.923        num_tors_div
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.9       0.000      0.000
2       -8.3       1.767      2.403
3       -7.9       1.304      1.862
4       -7.5       1.703      7.734
5       -7.5       1.399      7.739
6       -7.5       3.743      7.870
7       -7.4       5.115      7.452
8       -7.3       4.046      4.479
9       -7.2       3.710      7.817
Refine time 1.389
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.6       0.746      1.509
3       -8.4       1.856      2.758
4       -7.9       1.590      1.896
5       -7.8       1.328      7.758
6       -7.4       5.022      7.382
7       -7.3       4.211      4.844
8       -7.3       1.722      2.976
9       -7.1       4.628      5.311
Refine time 1.426
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.7       0.778      1.538
3       -8.1       1.824      2.475
4       -8.1       1.891      2.789
5       -7.8       1.248      7.721
6       -7.4       5.044      7.414
7       -7.3       1.736      2.862
8       -7.3       3.791      4.385
9       -7.2       4.074      4.473
Refine time 1.407
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.7       0.761      1.536
3       -8.2       1.922      2.861
4       -7.6       1.824      7.662
5       -7.5       1.264      7.764
6       -7.5       5.089      7.475
7       -7.1       4.032      4.531
8       -7.1       4.696      5.297
9       -7.0       4.807      8.489
Refine time 1.512
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -9.0       0.706      1.493
3       -8.3       0.895      1.047
4       -8.2       1.854      2.448
5       -7.6       1.841      2.158
6       -7.5       1.702      7.731
7       -7.3       5.143      7.462
8       -7.3       4.090      4.491
9       -7.3       2.025      7.771
Refine time 1.472
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.6       0.731      1.501
3       -8.5       1.897      2.763
4       -7.9       1.314      7.739
5       -7.6       1.597      7.705
6       -7.4       1.886      3.246
7       -7.3       5.037      7.432
8       -7.3       4.220      4.848
9       -7.0       4.669      5.276
Refine time 1.479
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -8.9       0.760      1.545
3       -8.2       1.854      2.470
4       -7.6       1.257      7.735
5       -7.6       1.666      7.682
6       -7.6       1.815      2.071
7       -7.5       1.307      1.761
8       -7.4       5.088      7.435
9       -7.4       3.999      4.430
Refine time 1.533
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.3       1.913      2.779
3       -8.2       1.800      2.405
4       -7.9       1.252      7.715
5       -7.5       1.790      7.630
6       -7.3       5.046      7.464
7       -7.3       3.886      4.271
8       -7.2       4.288      4.955
9       -7.0       4.614      5.208
Refine time 1.477
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.3       0.000      0.000
2       -7.9       1.175      2.179
3       -7.9       1.323      2.613
4       -7.5       1.462      7.712
5       -7.3       4.042      7.343
6       -7.2       3.445      7.113
7       -7.1       3.818      4.170
8       -7.1       4.071      7.165
9       -6.8       4.271      4.973
Refine time 1.407
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.4       0.000      0.000
2       -7.8       1.203      2.200
3       -7.6       1.460      7.754
4       -7.3       4.074      7.356
5       -7.1       1.560      7.670
6       -7.1       4.045      7.155
7       -7.0       3.802      4.178
8       -7.0       4.037      4.876
9       -6.9       4.130      4.579
Refine time 1.381
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.4       0.000      0.000
2       -7.8       1.423      7.728
3       -7.7       1.178      2.185
4       -7.4       3.985      7.250
5       -7.4       3.628      3.992
6       -7.4       1.435      7.732
7       -7.3       0.994      2.500
8       -7.1       3.963      7.046
9       -7.0       4.054      4.778
Refine time 1.378
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.4       0.000      0.000
2       -8.1       0.709      1.502
3       -7.8       1.178      2.188
4       -7.8       1.595      7.405
5       -7.5       4.030      7.318
6       -7.3       1.440      7.748
7       -7.3       3.750      4.100
8       -7.2       4.049      7.105
9       -7.1       4.144      7.445
Refine time 1.433
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.3       0.000      0.000
2       -7.9       1.343      2.635
3       -7.7       1.467      7.684
4       -7.7       3.380      3.779
5       -7.3       4.006      7.206
6       -7.1       3.997      7.020
7       -7.0       4.046      4.747
8       -6.9       4.525      4.906
9       -6.9       1.170      1.770
Refine time 1.385
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.1       0.000      0.000
2       -7.8       1.391      2.657
3       -7.8       1.569      7.406
4       -7.4       1.318      7.730
5       -7.3       4.070      7.263
6       -7.3       1.339      7.634
7       -7.2       3.605      3.988
8       -7.2       0.796      2.054
9       -7.1       1.324      7.720
Refine time 1.411
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.2       0.000      0.000
2       -8.1       0.760      1.556
3       -8.0       1.667      7.391
4       -7.9       1.355      2.639
5       -7.8       1.457      7.690
6       -7.7       3.397      3.798
7       -7.6       1.204      2.212
8       -7.4       4.019      7.208
9       -7.4       1.377      7.704
Refine time 1.431
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.2       0.000      0.000
2       -7.8       1.369      2.645
3       -7.7       0.767      1.536
4       -7.7       1.618      7.388
5       -7.6       1.500      7.681
6       -7.6       3.352      3.763
7       -7.4       1.550      7.440
8       -7.4       4.062      7.275
9       -7.2       4.045      7.055
Refine time 1.425
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.5       0.000      0.000
2       -8.0       1.459      1.773
3       -7.8       3.344      4.097
4       -7.8       3.507      4.866
5       -7.4       1.408      2.801
6       -7.4       4.231      7.281
7       -7.4       1.702      7.318
8       -7.3       1.699      7.490
9       -7.2       1.087      2.433
Refine time 1.468
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.5       0.000      0.000
2       -7.8       3.662      4.785
3       -7.8       3.546      5.008
4       -7.7       1.656      7.404
5       -7.6       2.221      7.070
6       -7.5       4.385      7.341
7       -7.4       1.010      2.377
8       -7.2       4.414      7.651
9       -7.1       4.397      7.326
Refine time 1.468
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.2       0.000      0.000
2       -7.8       3.440      4.793
3       -7.7       1.223      2.729
4       -7.7       3.584      4.423
5       -7.5       3.115      7.053
6       -7.4       1.704      7.364
7       -7.4       4.194      7.217
8       -7.3       4.350      7.598
9       -7.3       1.553      7.321
Refine time 1.504
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.6       0.000      0.000
2       -7.9       1.551      1.798
3       -7.9       3.268      3.931
4       -7.9       3.593      4.836
5       -7.9       1.666      7.192
6       -7.8       1.650      7.397
7       -7.5       4.207      7.160
8       -7.5       1.598      2.879
9       -7.2       4.243      7.426
Refine time 1.483
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.5       0.000      0.000
2       -7.9       3.122      3.813
3       -7.9       3.515      4.783
4       -7.8       3.571      4.759
5       -7.8       1.810      7.186
6       -7.6       1.381      1.646
7       -7.6       3.159      7.170
8       -7.5       4.279      7.219
9       -7.5       3.010      6.988
Refine time 1.485
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.4       0.000      0.000
2       -8.4       0.701      1.493
3       -8.4       1.140      2.238
4       -8.1       1.326      2.690
5       -7.5       1.502      7.635
6       -7.4       1.675      7.729
7       -7.3       4.908      7.475
8       -7.0       4.962      7.793
9       -6.9       1.342      1.770
Refine time 1.444
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.4       0.000      0.000
2       -8.2       1.331      2.674
3       -8.1       1.149      2.246
4       -7.7       1.352      7.728
5       -7.5       1.635      7.680
6       -7.3       4.912      7.590
7       -7.2       3.194      4.106
8       -7.0       4.977      7.894
9       -6.9       1.471      1.938
Refine time 1.442
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.6       0.700      1.515
3       -8.0       3.598      3.933
4       -7.9       0.874      2.093
5       -7.8       1.054      2.535
6       -7.7       3.932      4.324
7       -7.6       1.363      7.630
8       -7.4       3.867      7.187
9       -7.4       2.843      3.172
Refine time 1.404
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.5       1.334      2.626
3       -8.4       1.174      2.197
4       -8.2       3.473      3.826
5       -7.8       4.063      4.504
6       -7.7       1.450      7.672
7       -7.4       3.867      7.177
8       -7.4       4.296      4.960
9       -7.2       1.387      7.677
Refine time 1.425
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.3       3.413      3.763
3       -8.2       1.172      2.189
4       -7.9       4.093      4.526
5       -7.8       1.407      7.684
6       -7.6       1.299      1.882
7       -7.5       3.857      7.165
8       -7.4       4.260      4.927
9       -7.2       3.885      6.997
Refine time 1.418
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.4       1.320      2.611
3       -8.4       3.427      3.782
4       -8.2       1.183      2.203
5       -7.9       4.089      4.525
6       -7.6       1.604      1.849
7       -7.5       1.089      1.409
8       -7.5       3.861      7.192
9       -7.5       1.382      7.704
Refine time 1.407
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.9       0.000      0.000
2       -8.6       0.763      1.549
3       -8.3       1.355      2.626
4       -8.2       3.367      3.737
5       -7.7       3.903      4.245
6       -7.5       1.491      7.660
7       -7.5       3.952      7.249
8       -7.4       4.297      5.059
9       -7.3       1.096      2.278
Refine time 1.466
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.7       0.000      0.000
2       -8.6       0.767      1.539
3       -8.5       1.368      2.647
4       -8.3       3.353      3.728
5       -8.2       1.194      2.202
6       -7.8       3.800      4.151
7       -7.7       1.129      2.586
8       -7.5       3.899      7.192
9       -7.4       4.324      5.081
Refine time 1.406
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.6       3.367      3.724
3       -8.1       1.184      2.193
4       -7.9       4.221      4.547
5       -7.6       1.574      2.781
6       -7.5       1.376      7.676
7       -7.5       3.898      7.175
8       -7.4       4.148      4.807
9       -7.2       3.901      6.976
Refine time 1.466
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.9       0.000      0.000
2       -8.5       0.762      1.552
3       -8.5       3.376      3.727
4       -8.2       1.328      2.612
5       -7.9       0.902      2.106
6       -7.8       3.936      4.305
7       -7.4       4.122      4.798
8       -7.4       4.042      7.241
9       -7.2       4.612      7.472
Refine time 1.424
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.8       1.920      2.676
3       -8.1       2.000      3.142
4       -7.8       1.664      1.890
5       -7.5       5.015      7.415
6       -7.2       4.161      4.680
7       -7.2       4.649      7.716
8       -7.0       4.019      4.794
9       -7.0       5.240      7.714
Refine time 1.653
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -8.8       2.038      3.059
3       -8.8       1.918      2.700
4       -8.8       0.758      1.496
5       -7.9       1.670      1.871
6       -7.7       1.485      1.636
7       -7.6       5.109      7.602
8       -7.5       4.299      7.987
9       -7.3       2.596      7.883
Refine time 1.662
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
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```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -8.8       0.758      1.499
3       -8.7       1.939      2.713
4       -8.6       2.066      3.095
5       -7.5       4.971      7.368
6       -7.3       4.161      4.568
7       -7.2       3.989      7.908
8       -7.2       3.991      4.700
9       -7.0       2.662      7.885
Refine time 1.617
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.9       1.961      2.710
3       -8.7       2.067      3.086
4       -7.9       1.649      1.936
5       -7.4       2.739      8.013
6       -7.4       5.003      7.320
7       -7.3       4.112      4.580
8       -7.2       1.677      2.475
9       -7.0       2.696      7.837
Refine time 1.628
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.2       0.000      0.000
2       -8.8       1.941      2.701
3       -7.9       1.697      1.974
4       -7.4       5.029      7.367
5       -7.3       1.515      1.745
6       -7.3       4.112      4.582
7       -7.2       2.944      8.019
8       -7.2       2.679      7.810
9       -7.0       2.740      3.884
Refine time 1.654
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.0       0.000      0.000
2       -8.9       0.743      1.494
3       -8.8       1.954      2.706
4       -8.7       2.072      3.084
5       -7.5       2.560      7.828
6       -7.5       4.279      7.957
7       -7.4       5.095      7.532
8       -7.3       2.528      7.793
9       -7.2       3.982      4.421
Refine time 1.679
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -9.0       0.747      1.502
3       -8.8       1.943      2.698
4       -8.6       2.068      3.078
5       -7.5       5.128      7.597
6       -7.3       3.993      4.481
7       -7.2       2.505      7.815
8       -7.2       4.574      7.924
9       -7.0       2.838      3.359
Refine time 1.564
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.7       0.000      0.000
2       -8.6       0.701      1.460
3       -8.4       1.385      2.322
4       -7.9       1.487      2.744
5       -7.7       1.533      8.059
6       -7.7       4.204      5.159
7       -7.7       3.994      4.831
8       -7.6       4.537      7.710
9       -7.5       1.359      7.978
Refine time 1.538
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.7       0.000      0.000
2       -8.7       0.712      1.448
3       -8.5       1.525      2.728
4       -7.8       1.572      8.038
5       -7.7       4.270      5.247
6       -7.7       4.074      4.999
7       -7.6       4.517      7.769
8       -7.5       1.502      8.164
9       -7.3       3.450      4.105
Refine time 1.591
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.7       0.000      0.000
2       -8.5       1.519      2.706
3       -8.3       1.185      1.199
4       -8.2       1.359      8.084
5       -7.8       1.603      7.995
6       -7.6       4.059      4.857
7       -7.6       1.561      2.565
8       -7.4       4.456      7.668
9       -7.2       2.426      3.967
Refine time 1.561
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.7       0.000      0.000
2       -8.4       0.844      1.539
3       -8.3       1.590      2.790
4       -8.0       1.561      7.973
5       -7.9       1.263      8.086
6       -7.7       4.108      5.054
7       -7.5       4.469      7.724
8       -7.2       1.570      7.966
9       -7.1       1.755      2.085
Refine time 1.583
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.6       0.000      0.000
2       -8.4       0.813      1.533
3       -8.1       1.635      2.834
4       -7.9       1.291      8.071
5       -7.9       1.347      2.299
6       -7.7       4.062      5.008
7       -7.6       3.808      8.053
8       -7.6       4.522      7.741
9       -7.6       4.266      5.248
Refine time 1.588
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.7       0.000      0.000
2       -8.5       0.752      1.488
3       -8.4       1.492      2.697
4       -8.1       1.368      2.297
5       -7.9       1.288      8.077
6       -7.9       1.591      7.954
7       -7.7       1.689      7.965
8       -7.7       1.495      8.004
9       -7.6       3.823      8.019
Refine time 1.557
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.6       0.776      1.513
3       -8.2       1.274      8.047
4       -8.2       1.504      2.729
5       -8.1       1.323      2.294
6       -7.8       1.550      7.970
7       -7.7       1.503      7.962
8       -7.6       4.241      5.192
9       -7.5       4.479      7.655
Refine time 1.555
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -8.8       1.061      1.608
3       -8.7       1.357      2.349
4       -7.4       1.739      7.677
5       -7.3       1.941      2.221
6       -7.3       4.614      8.864
7       -7.3       1.415      7.751
8       -7.2       1.422      7.637
9       -7.1       3.996      4.549
Refine time 1.451
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.2       0.000      0.000
2       -8.6       1.367      2.292
3       -7.6       1.947      2.068
4       -7.3       1.444      7.651
5       -7.3       3.255      4.188
6       -7.3       1.766      7.717
7       -7.2       1.202      7.672
8       -7.1       3.588      7.651
9       -7.1       4.662      5.552
Refine time 1.438
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.5       0.000      0.000
2       -9.2       1.349      2.347
3       -8.9       0.974      1.525
4       -8.6       1.345      1.517
5       -7.7       1.457      2.643
6       -7.7       1.146      7.724
7       -7.4       1.746      7.731
8       -7.2       3.741      7.649
9       -7.1       4.012      4.603
Refine time 1.387
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.3       0.000      0.000
2       -8.8       1.024      1.567
3       -8.7       1.351      2.349
4       -7.9       3.524      7.626
5       -7.3       4.711      9.001
6       -7.1       4.726      5.554
7       -7.1       1.849      7.751
8       -7.1       4.433      7.801
9       -7.0       4.145      4.809
Refine time 1.397
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.6       0.000      0.000
2       -8.8       1.055      1.597
3       -8.8       1.389      2.305
4       -8.4       1.729      2.793
5       -8.0       1.215      7.767
6       -7.4       1.421      7.671
7       -7.3       1.861      2.353
8       -7.2       3.998      4.559
9       -7.1       1.748      7.666
Refine time 1.407
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.1       0.000      0.000
2       -8.8       1.050      1.604
3       -8.7       1.401      2.353
4       -8.3       1.787      2.882
5       -7.7       2.923      7.599
6       -7.3       4.729      8.882
7       -7.2       1.813      7.589
8       -7.1       1.478      7.664
9       -7.1       4.727      5.558
Refine time 1.388
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.4       0.000      0.000
2       -8.7       1.369      2.322
3       -8.0       1.130      1.145
4       -7.8       2.847      7.609
5       -7.6       1.965      2.082
6       -7.2       2.126      7.363
7       -7.2       4.735      5.594
8       -7.2       4.687      8.904
9       -7.1       1.455      7.677
Refine time 1.448
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -9.4       0.000      0.000
2       -9.2       1.390      2.340
3       -8.8       0.963      1.517
4       -8.0       1.153      7.743
5       -7.4       1.753      7.684
6       -7.4       1.437      2.730
7       -7.3       3.342      3.521
8       -7.2       1.530      2.259
9       -7.2       4.661      8.527
Refine time 1.394
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.9       0.000      0.000
2       -8.6       1.466      2.689
3       -8.3       1.381      2.365
4       -7.9       1.303      7.762
5       -7.5       4.846      7.400
6       -7.2       4.334      5.047
7       -7.2       4.701      8.824
8       -7.1       4.283      5.055
9       -7.0       4.591      8.482
Refine time 1.443
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.7       0.693      1.485
3       -8.5       1.478      2.700
4       -8.2       1.337      2.276
5       -7.7       1.607      1.873
6       -7.4       4.839      7.387
7       -7.3       1.548      7.733
8       -7.2       4.355      5.064
9       -7.1       4.329      5.125
Refine time 1.415
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.7       0.701      1.487
3       -8.6       1.512      2.711
4       -7.9       1.283      7.765
5       -7.6       1.471      2.373
6       -7.4       4.884      7.426
7       -7.2       4.324      4.999
8       -7.0       4.288      5.022
9       -6.9       4.581      8.208
Refine time 1.485
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.6       0.773      1.533
3       -8.4       1.511      2.761
4       -7.6       1.570      7.677
5       -7.5       1.683      7.744
6       -7.5       4.861      7.455
7       -7.4       3.907      7.926
8       -7.3       4.250      4.971
9       -7.1       1.523      7.696
Refine time 1.449
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.6       0.772      1.536
3       -8.3       1.340      2.326
4       -8.2       1.497      2.714
5       -7.5       1.547      7.648
6       -7.4       3.507      4.276
7       -7.4       4.861      7.415
8       -7.2       4.291      4.980
9       -7.2       4.322      5.115
Refine time 1.494
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.4       1.539      2.753
3       -8.4       1.395      2.337
4       -7.6       1.249      7.744
5       -7.5       1.561      7.645
6       -7.3       4.926      7.458
7       -7.1       4.800      5.673
8       -6.9       2.045      7.794
9       -6.9       1.810      7.837
Refine time 1.437
Using random seed: 0
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
```

```
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
1       -8.8       0.000      0.000
2       -8.3       1.551      2.798
3       -8.2       1.337      2.262
4       -7.7       1.463      7.659
5       -7.4       4.922      7.408
6       -7.4       1.652      2.048
7       -7.4       1.730      7.758
8       -7.3       4.161      4.690
9       -7.2       1.748      7.752
Refine time 1.536
Loop time 100.964

smina finished with exit code 0. Output: Trial10_pH7.4_docked.sdf
Log: Trial10_pH7.4_docked.log
Run record: Trial10_pH7.4_docked.run.json

✔ Coverage: all 9 ligands have poses.
Note: 18 of 540 poses could not be read as written by smina and were rebuilt from the submitted ligand (coordinates
kept, bond orders from the input).

✔ Box check: no pose within 1 Å of the box edge.
```

5. View docking results

Poses are sorted by smina score (best first). Step through them with the slider; the ligand from the original PDB (reference) is shown with red carbons and the current pose with green carbons. Untick **Ligand from PDB (red)** to hide the reference ligand.

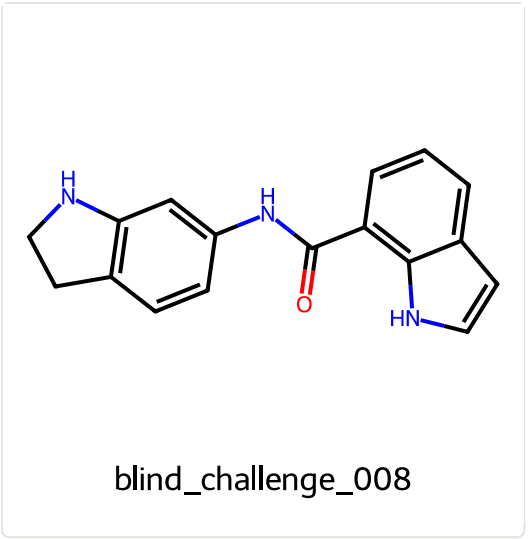
After **flexible docking** each pose is shown together with *its own* flexed side chains (magenta carbons), with the original crystal side-chain positions overlaid in thin grey (toggle). The receptor for each pose is rebuilt from the rigid protein plus that pose's model in `<output>_flexres.pdb`.

In [13]:
```python
# Re-run this cell after smina has finished
DOCKED_SDF = Path(panel.output.value)
poses, pose_table, FLEX_PDB = [], None, None
if DOCKED_SDF.exists():
    poses = dt.load_poses(DOCKED_SDF)
    FLEX_PDB = dt.flex_file_for(DOCKED_SDF)       # present only after flexible docking
    FLEX_PDB = FLEX_PDB if FLEX_PDB.exists() else None
    print(f"{len(poses)} poses in {DOCKED_SDF}")
    if FLEX_PDB:
        print(f"Flexible side chains: {FLEX_PDB}")
    pose_table = dt.results_browser(poses, site.protein, site.ligand_ref, flex_pdb=FLEX_PDB)
else:
    print(f"{DOCKED_SDF} not found - press 'Run smina' above, then re-run this cell")
```

```
Note: 18 of 540 poses could not be read as written by smina and were rebuilt from the submitted ligand (coordinates
kept, bond orders from the input).
540 poses in Trial10_pH7.4_docked.sdf
```

1/540 **blind_challenge_008** (pose_441) smina affinity: **-9.55** kcal/mol

Protein (RasMol secondary structure): helix sheet coil/turn



In [14]:
```
pose_table.head(10) if pose_table is not None else None
```

Out[14]:

	pose_id	name	minimizedAffinity
0	pose_441	blind_challenge_008	-9.55072
1	pose_423	blind_challenge_008	-9.52739
2	pose_468	blind_challenge_008	-9.41364
3	pose_459	blind_challenge_008	-9.40637
4	pose_432	blind_challenge_008	-9.31476
5	pose_424	blind_challenge_008	-9.23334
6	pose_469	blind_challenge_008	-9.22708
7	pose_414	blind_challenge_008	-9.15423
8	pose_315	blind_challenge_006	-9.15078
9	pose_405	blind_challenge_008	-9.13256

6. Binding-affinity prediction with AEV-PLIG

AEV-PLIG is a graph neural network scoring function (atomic environment vectors + protein–ligand interaction graphs). The model used is `model_GATv2Net_ligsim90_fep_benchmark` (ensemble of 10), copied with the prediction code into `aev_plig/` in this project. Predictions are **pK** values (higher = tighter binding); the spread across the 10 models is reported as `AEV_PLIG_pK_std`.

> **Metal-containing binding sites.** AEV-PLIG reads only the standard amino acids, so metal ions (e.g. Zn, Mg), waters and cofactors are ignored. On its CASF-2016 test set, complexes with a metal coordinating the ligand were predicted less accurately (RMSE 1.44 vs 1.00 pK units). smina has no metal-specific term either. Treat scores for metalloenzymes with caution; see the README section *Metal-containing binding sites*. This ZIKV NS2B-NS3 protease structure has **no metal** in the binding site, so that caveat does not apply here.

All results are exported to `predictions/` next to the docked SDF:

File	Content
`poses/<docked>/pose_<i>.sdf`	one file per pose (AEV-PLIG input)
`poses/<docked>/pose_<i>_receptor.pdb`	flexible docking only: the receptor with that pose's flexed side chains (each pose is scored against its own receptor)
`<docked>_aevplig_input.csv`	AEV-PLIG input table
`<docked>_predictions.csv`	raw AEV-PLIG output (10 models + mean)
`<docked>_summary.csv`	name, pose, smina score, pK, pK std, best-pose flags
`<docked>_scored.sdf`	all poses annotated with pK
`<docked>_best_poses.sdf`	highest-pK pose per ligand
`<docked>_best_poses_table.csv`	best pose per ligand **plus the ligand from the original PDB** in its crystal pose: AEV-PLIG pK (also the red dashed line on the plot) and smina scores, the **box check** (`box_margin` , `at_box_edge`), plus **ligand efficiency** (`heavy_atoms` , `LE_AEV_PLIG` = 1.364 × pK / heavy atoms, `LE_smina` = −smina score / heavy atoms; kcal/mol per heavy atom) – `minimizedAffinity` after local minimisation in the site (comparable with the docked poses), `smina_score_only` for the unmodified crystal coordinates and `minimizedRMSD` (how far minimisation moved it)
`pdb_ligand/`	the PDB ligand as scored by smina, its minimised pose and the smina log
`<docked>_best_poses.pse`	PyMOL session of the best poses (section 7)
`best_poses_pdb/<docked>/<ligand>_<pose>_ligand.pdb` and `..._complex.pdb`	highest-pK pose per ligand in PDB format: ligand only, and protein–ligand complex (ligand = chain L, resname LIG) using that pose's own receptor (flexed side chains after flexible docking); REMARKs give smina score and pK
`<docked>_smina_vs_aevplig.png`	score comparison plot
copies of docked SDF, receptor, `_flexres.pdb` (flexible), reference ligand, conformers and images	for a self-contained record

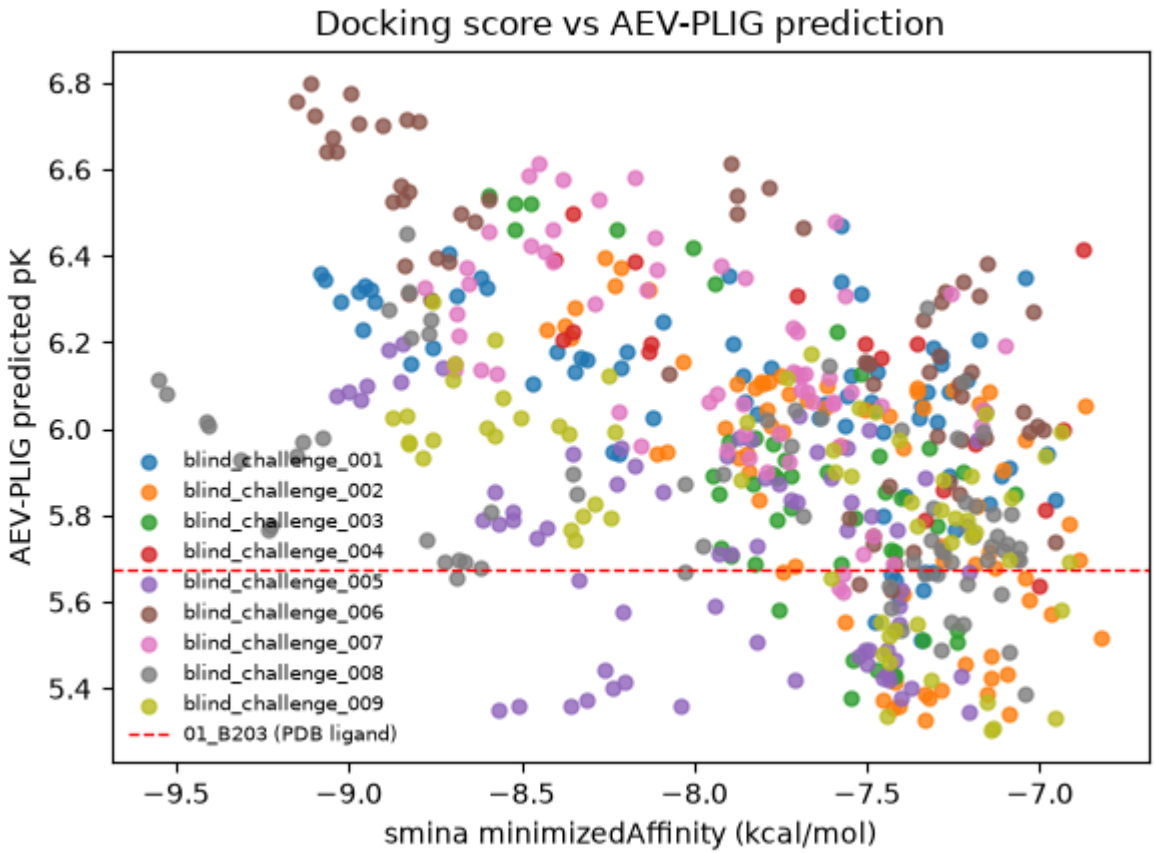
In [15]:

```
predict_button = dt.widgets.Button(description="Run AEV-PLIG", button_style="success", icon="play")
predict_out = dt.widgets.Output()
summary = None

def _predict(_=None):
    global summary
    predict_out.clear_output()
    with predict_out:
        if not DOCKED_SDF.exists():
            print("No docked SDF yet - run smina and the 'View docking results' cell first")
            return
        extras = [CONF_SDF, site.ligand_pdb, site.ligand_ref, complex_png, site_png,
                  DOCKED_SDF.with_suffix(".log")]
        # smina scores for the ligand from the PDB in its crystal pose (same smina + scoring function)
        ref_smina = None
        if REF_MOL is not None:
            ref_smina = dt.smina_score_ligand(REF_MOL, site.protein,
                                              DOCKED_SDF.parent / "predictions" / "pdb_ligand",
                                              smina=panel.smina.value, scoring=panel.scoring_arg)
        summary = dt.run_aev_plig(DOCKED_SDF, site.protein, extra_files=extras, flex_pdb=FLEX_PDB,
                                  reference_mol=REF_MOL,     # also score the ligand from the PDB
                                  reference_smina=ref_smina)
        display(summary.head(10))

predict_button.on_click(_predict)
display(dt.widgets.VBox([predict_button, predict_out]))
if AUTO_RUN:
    _predict()
```

```
[20:21:35] Warning: molecule is tagged as 2D, but at least one Z coordinate is not zero. Marking the mol as 3D.
PDB ligand smina: score_only -6.91, minimized -8.26 kcal/mol (RMSD 0.80 Å)
Note: 18 of 540 poses could not be read as written by smina and were rebuilt from the submitted ligand (coordinates
kept, bond orders from the input).
$ python process_and_predict.py --dataset_csv=predictions/Trial10_pH7.4_docked_aevplig_input.csv --
data_name=Trial10_pH7.4_docked --trained_model_name=model_GATv2Net_ligsim90_fep_benchmark --num_workers=24 --
output_dir=predictions
PDB ligand 01_B203: AEV-PLIG pK 5.67 +/- 0.24
Best pose per ligand exported as PDB (9 ligands) to predictions/best_poses_pdb/Trial10_pH7.4_docked
```




```
All results exported to predictions
```

	pose_id	name	minimizedAffinity	AEV_PLIG_pK	AEV_PLIG_pK_std	heavy_atoms	LE_AEV_PLIG	LE_smina	box_margin	at_box_edge	...	preds_0	preds_1	preds_2	preds_3	preds_4	preds_5	preds_6	preds_7	preds_8	preds
0	pose_333	blind_challenge_006	-9.11234	6.797267	0.400942	22	0.421506	0.414197	3.6896	False	...	7.003190	6.449265	7.070715	6.554767	7.100150	6.274014	7.517409	6.324935	6.699533	6.9786
1	pose_334	blind_challenge_006	-8.99596	6.776013	0.517688	22	0.420188	0.408907	3.8827	False	...	7.558510	6.202136	7.010823	6.279401	6.992015	6.551013	7.123636	6.038880	6.611279	7.3924
2	pose_315	blind_challenge_006	-9.15078	6.755940	0.421402	22	0.418943	0.415945	3.9663	False	...	7.471357	6.320669	7.010229	6.454211	6.819148	6.516921	6.825831	6.301996	6.463636	7.3753
3	pose_288	blind_challenge_006	-9.10116	6.727074	0.419719	22	0.417153	0.413689	3.6761	False	...	7.149402	6.521333	7.033169	6.458802	7.048016	6.446998	7.262666	5.925834	6.502030	6.9224
4	pose_291	blind_challenge_006	-8.83585	6.716607	0.535075	22	0.416504	0.401630	3.9305	False	...	7.533874	6.201456	6.800894	6.270289	7.027946	6.514411	7.076098	5.884932	6.487088	7.3690
5	pose_298	blind_challenge_006	-8.79761	6.710677	0.490927	22	0.416136	0.399891	4.0549	False	...	7.393337	6.194388	7.309625	6.425050	6.669565	6.568229	6.661405	6.118748	6.353186	7.4132
6	pose_306	blind_challenge_006	-8.97514	6.705144	0.463779	22	0.415793	0.407961	4.0206	False	...	7.347027	6.028910	7.348566	6.418143	6.741957	6.510499	6.660876	6.214677	6.558910	7.2218
7	pose_325	blind_challenge_006	-8.90116	6.702274	0.522775	22	0.415615	0.404598	3.9214	False	...	7.316704	5.987438	7.380123	6.434479	6.923917	6.447163	6.740887	5.868718	6.743873	7.1794
8	pose_324	blind_challenge_006	-9.04558	6.676189	0.465821	22	0.413998	0.411163	3.7376	False	...	6.874043	6.097491	7.333731	6.509044	7.023315	6.101252	7.132371	6.049394	6.709217	6.9320
9	pose_297	blind_challenge_006	-9.06545	6.641858	0.463427	22	0.411869	0.412066	3.8057	False	...	6.842058	6.278629	7.256794	6.409515	6.916008	6.291328	7.075273	5.747883	6.584549	7.0165

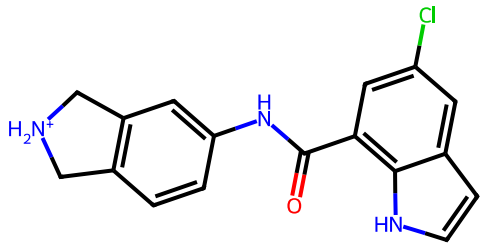
10 rows × 23 columns

Step through poses ranked by AEV-PLIG pK

In [16]:
```
if summary is not None:
    dt.results_browser(poses, site.protein, site.ligand_ref, predictions=summary, sort_by="AEV_PLIG_pK",
                       flex_pdb=FLEX_PDB)
```

1/540 **blind_challenge_006** (pose_333) smina affinity: **-9.11** kcal/mol AEV-PLIG pK: **6.80**

Protein (RasMol secondary structure): helix sheet coil/turn



blind_challenge_006

In [17]:
```
# Best pose per ligand by AEV-PLIG, plus the ligand from the original PDB in its crystal pose.
# Ligand efficiency (kcal/mol per heavy atom): LE_AEV_PLIG = 1.364 x pK / heavy atoms; LE_smina = -smina score /
heavy atoms
# (best poses are also exported as PDB: ligand only + protein-ligand complex)
if summary is not None:
    best_table = pd.read_csv(DOCKED_SDF.parent / "predictions" / f"{DOCKED_SDF.stem}_best_poses_table.csv")
    display(best_table.style.format({"minimizedAffinity": "{:.2f}", "AEV_PLIG_pK": "{:.2f}",
                                     "AEV_PLIG_pK_std": "{:.2f}", "smina_score_only": "{:.2f}",
                                     "minimizedRMSD": "{:.2f}", "LE_AEV_PLIG": "{:.3f}",
                                     "LE_smina": "{:.3f}", "box_margin": "{:.2f}"}, na_rep="–")
            .apply(lambda r: ["background-color: #fde2e2" if r["source"] == "ligand from PDB" else ""] * len(r),
                   axis=1).hide(axis="index"))
    best_pdb_dir = DOCKED_SDF.parent / "predictions" / "best_poses_pdb" / DOCKED_SDF.stem
    print(f"PDB files in {best_pdb_dir}:")
    for f in sorted(best_pdb_dir.glob("*.pdb")):
        print("  ", f.name)
```

source	name	pose_id	minimizedAffinity	AEV_PLIG_pK	AEV_PLIG_pK_std	heavy_atoms	LE_AEV_PLIG	LE_smina	box_margin	at_box_edge	smina_interaction	smina_score_only	minimizedRMSD
docked (best AEV-PLIG pose)	blind_challenge_006	pose_333	-9.11	6.80	0.40	22	0.422	0.414	3.69	False	-9.112120	–	–
docked (best AEV-PLIG pose)	blind_challenge_007	pose_361	-8.45	6.61	0.39	22	0.410	0.384	3.16	False	-8.480360	–	–
docked (best AEV-PLIG pose)	blind_challenge_003	pose_171	-8.60	6.54	0.41	22	0.406	0.391	3.95	False	-8.595090	–	–
docked (best AEV-PLIG pose)	blind_challenge_004	pose_191	-8.35	6.50	0.36	21	0.422	0.398	3.77	False	-8.414290	–	–
docked (best AEV-PLIG pose)	blind_challenge_001	pose_40	-7.58	6.47	0.41	21	0.420	0.361	3.53	False	-7.724330	–	–
docked (best AEV-PLIG pose)	blind_challenge_008	pose_442	-8.84	6.45	0.35	21	0.419	0.421	3.76	False	-8.817070	–	–
docked (best AEV-PLIG pose)	blind_challenge_002	pose_108	-8.26	6.40	0.48	21	0.416	0.393	4.19	False	-8.257770	–	–
docked (best AEV-PLIG pose)	blind_challenge_009	pose_531	-8.76	6.30	0.46	21	0.409	0.417	3.87	False	-8.756530	–	–
docked (best AEV-PLIG pose)	blind_challenge_005	pose_261	-8.84	6.20	0.47	21	0.403	0.421	4.26	False	-8.844180	–	–
ligand from PDB	01_B203	crystal pose	-8.26	5.67	0.24	29	0.267	0.285	4.00	False	-8.260930	-6.91	0.80

```
PDB files in predictions/best_poses_pdb/Trial10_pH7.4_docked:
   blind_challenge_001_pose_40_complex.pdb
   blind_challenge_001_pose_40_ligand.pdb
   blind_challenge_002_pose_108_complex.pdb
   blind_challenge_002_pose_108_ligand.pdb
   blind_challenge_003_pose_171_complex.pdb
   blind_challenge_003_pose_171_ligand.pdb
   blind_challenge_004_pose_191_complex.pdb
   blind_challenge_004_pose_191_ligand.pdb
   blind_challenge_005_pose_261_complex.pdb
   blind_challenge_005_pose_261_ligand.pdb
   blind_challenge_006_pose_333_complex.pdb
   blind_challenge_006_pose_333_ligand.pdb
   blind_challenge_007_pose_361_complex.pdb
   blind_challenge_007_pose_361_ligand.pdb
   blind_challenge_008_pose_442_complex.pdb
   blind_challenge_008_pose_442_ligand.pdb
   blind_challenge_009_pose_531_complex.pdb
   blind_challenge_009_pose_531_ligand.pdb
```

7. Export best poses as a PyMOL session

Saves `predictions/<docked>_best_poses.pse` with the best AEV-PLIG pose of each ligand, ready to open in PyMOL:

- `receptor` – cartoon (RasMol secondary-structure colours) with binding-site side chains as light-blue sticks
- one group per ligand, ranked by pK (`01_<ligand>` , `02_<ligand>` ...) containing the pose (`_pose`), its polar contacts (`_hbonds`) and, after flexible docking, that pose's flexed side chains (`_flexed` , magenta)
- `PDB_ligand` – the ligand from the original PDB (red carbons)

Only the top-ranked ligand is switched on initially (tick **Show all ligands** to enable every one); toggle groups in PyMOL's object panel. Each pose's title shows its name, pose id, pK and smina score.

In [18]:
```
pse_all = dt.widgets.Checkbox(value=False, description="Show all ligands when the session opens")
pse_button = dt.widgets.Button(description="Export PyMOL session", button_style="success", icon="download")
pse_out = dt.widgets.Output()

def _export_pse(_=None):
    pse_out.clear_output()
    with pse_out:
        if summary is None:
            print("Run AEV-PLIG first (section 6)")
            return
        dt.export_pymol_session(DOCKED_SDF, site.protein, reference_ligand=site.ligand_ref,
                                enable="all" if pse_all.value else "top")

pse_button.on_click(_export_pse)
display(dt.widgets.VBox([pse_all, pse_button, pse_out]))
if AUTO_RUN:
    _export_pse()
```

```
Note: 18 of 540 poses could not be read as written by smina and were rebuilt from the submitted ligand (coordinates
kept, bond orders from the input).
PyMOL session with 9 best poses written to predictions/Trial10_pH7.4_docked_best_poses.pse
```

In [19]:
```
```